RESEARCH RATIONALE & GROUP PLAN

LG TV Purchase Conversion

A focused comparison with Samsung in online retail

Working direction

Investigate how observed TV product views translate into purchases, then identify the next evidence or test LG should prioritise.

1. How the project took shape

The project began with the LG Electronics Track and purchase conversion as a possible route. We compared the behavioural and review data before narrowing the category, comparator and managerial decision. The logic remains business problem, focused research question, data, course method, insight and action. This revision checks the proposed methods against both supplied CSVs and the actual lessons. [G; D; A]

The working scope remains **televisions, LG versus Samsung, October-November 2019**, using the behavioural dataset. One substantive course method, supported by relevant descriptive analysis, can be sufficient. Using both CSVs or every method taught is not a requirement. The existing eight-page structure and 23 numbered sections are retained; the method choices are integrated into the argument. [G, p. 1]

2. Research question

Primary question

In the television category, how does LG's observed view-to-purchase conversion probability compare with Samsung's after accounting for initial product price and time?

Supporting diagnostic: Which stage of the observed purchase journey warrants closer investigation? Journey measures support the same managerial decision; they are not a separate project.

3. Managerial purpose and decision status

The purpose is to decide where further investigation or a retailer-supported test would be useful. A lower observed conversion rate is a symptom worth examining. It does not identify a pricing error, weak product information or checkout failure. Those explanations require variables the current files do not contain.

The fresh audit reproduces the original baseline but also shows that the size of the difference is sensitive to the product mix. The project should therefore establish the limits of the brand comparison before suggesting an intervention. Group agreement and instructor feedback are still needed. A smaller or inconclusive difference remains a valid result.

Internal planning guide, not the assessed final report. Historical results do not describe current LG performance. [A] denotes the fresh raw-data audit for this revision; course and project source keys are on page 8.

CHOOSING THE SCOPE

Why TV, and why Samsung?

4. The criteria behind the choice

TV conversion offers a measurable outcome, a focused comparator and a direct fit with binary response models. The reconstructed LG-Samsung cohort has **74,712 product-consideration rows and 2,150 purchases**. It covers 35,871 distinct users and 196 products. Selection was exploratory: sample size and feasibility support the choice, but a large sample does not remove selection bias. [A]

5. Why Samsung is the primary benchmark

The median initial price is 460.84 for LG and 493.96 for Samsung, in unidentified store currency units. The central distributions also overlap, giving a better basis for considering a price adjustment than a median comparison alone. Similar prices still do not establish equivalent screen sizes, technology, specifications or availability.

Initial-price quantile	LG	Samsung
25th percentile	390.86	381.74
Median	460.84	493.96
75th percentile	666.27	635.25

An exploratory restriction to the intersection of each brand's 1st-99th percentile intervals, **282.63-2,547.75**, retains 71,232 rows. This is a sensitivity sample, not a course rule or proof of product matching. More importantly, the five most-viewed products account for **42.3% of LG rows and 52.4% of Samsung rows**. The model must be interpreted as a comparison of these observed assortments, not pure brand preference. [A]

6. Washing machines: the closest category alternative

Comparable cohort	TV	Washing machines
LG rows / purchases	22,213 / 505	21,959 / 544
Samsung rows / purchases	52,499 / 1,645	32,684 / 910
LG / Samsung conversion	2.27% / 3.13%	2.48% / 2.78%

Washing machines remain credible: LG has slightly more purchase outcomes there. TV is retained because the joint cohort is larger and the analytical route is already developed. The smaller washing-machine gap does not make that question less valuable, and no equivalent model review establishes TV as inherently superior. Revisit the category for a substantive comparability problem or group preference, not to preserve significance. [A; R]

All displayed counts and price summaries on this page were recomputed from electronics_market_behavior.csv. Each row is a user-session-product consideration with an eligible first view.

ALTERNATIVES CONSIDERED

Why not segmentation or reviews?

7. Behavioural segmentation: a different targeting question

The event file could support user-level summaries of browsing breadth, activity and observed purchases. K-means, taught in Session 8 and practised in Session 9, would be appropriate if the decision were which visitor profiles to target. It would require meaningful features, scaling, a justified cluster count and a different feasible action for each profile. [C8; N9]

For the current RQ, clustering does not directly estimate the LG-Samsung conversion difference. Purchase-based clusters followed by comparisons of their purchase rates would also be partly mechanical. Two months cannot establish loyalty or lifetime value, and viewing LG does not make someone an LG-owned customer. Keep segmentation as a different project direction.

PCA and perceptual maps, taught in Sessions 10-11, also lack a suitable measurement basis here. The behavioural CSV contains eight columns: identifiers, event type/time, brand, category and price. It contains no consumer attribute-rating battery. A price-conversion chart can describe this sample, but should not be called a perceptual map. [C10; N11; A]

8. Negative reviews: a separate satisfaction project

The second file has **20,004 reviews of 2,750 products**, including 2,923 LG and 2,684 Samsung reviews. Its six categories are refrigerator/freezer, laundry, cooking, other appliances, dishwasher and air conditioner. There is no TV category, and its product IDs have **zero overlap** with the behavioural file. The dictionary explicitly rules out a join. These reviews cannot explain the observed TV conversion gap. [D; A]

A separate question about negative-review incidence could use binary Logit after defining a negative rating and accounting for category and period. Yet sampling caps, self-selection and repeated reviews per product matter. In the supplied LG sample, one- or two-star reviews represent 14.68% of verified reviews and 68.12% of unverified reviews. Unverified does not mean fake; neither figure is a population defect rate.

9. Why pricing is a control, not the project objective

Session 16 and the Session 17 case use demand quantities, prices, promotions and contribution calculations. Here, price is observed at an event, with no quantity field, promotion indicator, costs, specifications or availability. There is price variation within 146 of the 196 TV products, but it is not randomised variation. It does not identify an elasticity or an optimal discount. [C16; N17; A]

Using log(initial price) in a binary Logit adjusts a conversion association; its coefficient is not the log-log demand elasticity taught in pricing. Review prices are missing in 15,883 rows (79.4%), so the second file does not solve this problem. No added pricing model is justified for the present decision.

DATA & DEFINITIONS

What exactly are we measuring?

10. The evidence base and its boundaries

electronics_market_behavior.csv contains 649,605 events and eight columns over 1 October-30 November 2019. Each raw row records a view, cart addition or purchase, together with user, session, product, category, brand and price. It is not a customer record or complete order. The instructor curated brands and categories and sampled whole sessions. [D; A]

The retailer, country and currency are unidentified. This is a historical market-behaviour proxy, not proprietary LG data or representative total-market traffic. The review file describes different products and users. Its presence does not require the group to combine it with this analysis.

11. Analytical unit and purchase outcome

Use one row per **user_id x session_id x product_id** with at least one view. Retain the first view and its price/date. The fresh audit confirms that four session IDs occur under multiple users, so session_id alone is not a safe key. [A]

Define **Y = 1** if a purchase of that product occurs strictly after the first view in the same composite key; otherwise **Y = 0**. A zero means no qualifying purchase observed here, not no purchase elsewhere, in another session or beyond the data window. One user can contribute several considerations.

The TV subset has 2,331 raw purchase events; 19 lack any view in their composite key and 14 precede the first view. These are event counts, not additional purchase outcomes to add to the cohort. With repeated events collapsed to presence, the eligible cohort yields 2,150 purchase outcomes. Audit unusual sequences and the end of the observation window consistently for both brands.

12. The three journey indicators

Indicator	Numerator	Denominator
View-to-purchase	Rows with purchase after first view	All eligible viewed rows
View-to-cart	Rows with cart after first view	All eligible viewed rows
Cart-to-purchase	Rows with purchase after first eligible cart	Viewed rows with an eligible cart

These KPIs apply the measurement and journey logic of Sessions 3 and 5 and the Session 6 notebook. Not every qualifying purchase contains an intermediate cart event, so the two stage rates need not multiply to total conversion. Their different denominators also prevent treating stage gaps as causal contributions. [C3; C5; N6]

Raw-file audit: 1,247 exact duplicate events; no missing behavioural values or non-positive prices; product-brand/category mappings are consistent. Duplicate rows do not change these presence/first-view counts. Alternative timing rules and period-boundary sensitivity remain open. [A]

PRELIMINARY EVIDENCE

What the data now show

The fresh audit reconstructs the original definitions and reproduces the descriptive TV results. These are consideration-level sample rates, not causal brand effects or estimates of current market performance. [A]

TV cohort indicator	LG	Samsung
Eligible user-session-product rows	22,213	52,499
Purchase after first view	505	1,645
View-to-purchase	2.27%	3.13%
Cart after first view	1,073	3,311
View-to-cart	4.83%	6.31%
Purchase after eligible cart	374	1,330
Cart-to-purchase	34.86%	40.17%
Distinct products	105	91

13. How to interpret the descriptive comparison

The raw LG-minus-Samsung conversion gap is **-0.86 percentage points**. It cannot be converted into lost sales without a justified causal counterfactual. LG is lower in both months: October 2.70% versus 3.77%; November 2.09% versus 2.79%. Two months do not establish annual stability.

View-to-cart and cart-to-purchase differences suggest places to investigate. They do not identify defective pages or checkout friction. The cart comparison concerns a selected subset; its larger numerical gap does not show that checkout explains more of the overall difference.

14. The adjusted comparison and its uncertainty

The refitted Logit with log(initial price) and a November indicator gives standardised probabilities of **2.295% for LG and 3.121% for Samsung**, a **-0.827-point** difference. The LG odds ratio is 0.729. Standardisation compares brands over the same observed price/month distribution. [A; C12; N15]

This revision also computes the previously missing probability-gap interval. A delta-method 95% interval using **user-clustered** covariance is **[-1.107, -0.546] points**. Using **product-clustered** covariance instead gives **[-1.475, -0.179]**. The wider interval shows why product dependence deserves attention; the two separate calculations do not jointly address both dependence structures.

Intervals are approximate and conditional on the observed standardisation covariates. They do not cover omitted-variable bias or exploratory category selection. A final inference choice should be discussed with the instructor. Fresh audit [A], 25 October 2026; baseline point estimates agree with the supplied rationale.

ANALYTICAL PLAN

Methods chosen for these data

15. A focused sequence, not a collection of models

Binary Logit is the main course method because each eligible consideration has a yes/no purchase outcome. Begin with M1: brand only. M2 adds log(initial price) and November on the same cohort. Samsung is the reference. Sessions 12 and 15 provide the model, odds-ratio interpretation and predicted probabilities. Report the discrete LG-Samsung probability contrast, not only the coefficient. [C12; N15]

The outcome is not an exclusive choice between brands: 6,364 users have eligible views of both. Multinomial Logit is taught in the same lessons, but would require a new choice occasion, alternatives and no-purchase/multiple-purchase rules. It does not fit the present row definition merely because two brands are observed.

16. Checks that the audit makes worth prioritising

Model or sensitivity	Rows	LG-Samsung gap
M2 Logit: price + month	74,712	-0.827 pp
Same controls, linear probability model	74,712	-0.822 pp
M2 within central price overlap	71,232	-0.885 pp
M2 without each brand's most-viewed product	62,883	-0.431 pp

Functional form. The LPM taught in Session 12 is useful as a supporting comparison: its estimate is similar and none of its observed fitted values leaves [0,1]. Keep Logit as the main model and the LPM as a compact check. This agreement does not validate causality.

Product mix. Removing the single product with the most eligible rows in each brand nearly halves the gap. The rule uses volume, not conversion; it changes the sample and standardisation population. This does not invalidate the baseline, but makes concentration and missing specifications more important than adding more techniques. The price-overlap restriction alone changes the gap much less. [A]

Interactions. An exploratory LG x log(price) term, motivated by Session 7, produces virtually the same average gap and no clear improvement in fit (AIC 19,339.7 versus 19,337.7). Retain the simpler main specification. This is not proof that price associations are identical; a new interaction needs a substantive reason and suitable support.

17. What the analysis must deliver

Produce the cohort audit, clear journey measures, M1/M2 probabilities and a small set of informative sensitivities. Prioritise repeated-user/product dependence, concentration, timing and period boundaries. Use information known at the first view: later carts, final spending and whole-session counts are unsuitable starting-point controls. Product fixed effects absorb brand when each product has one brand.

Course anchors: Session12_Logit.pdf, pp. 7-22; session15_response_models.ipynb, sections 1-3; Session7_OLS_Extention_Application.pdf, pp. 5-13. Pseudo-R-squared is not OLS variance explained; fit improvements do not establish causality. All table results were computed in [A].

MANAGERIAL IMPLICATIONS

Strategies depend on what survives scrutiny

The intended chain remains evidence, plausible explanation, missing information, targeted test and decision. The raw-data checks strengthen the case for investigating **assortment and product concentration** before treating the average difference as a general LG weakness. The following are conditional strategies, not established failures.

18. If the view-to-cart gap remains informative

Investigate what visitors see before adding a TV to the cart: specifications, comparison information, commercial terms, stock and traffic source. The event file records none of those explanations directly. Use the product concentration results to identify where a content or assortment audit would be most informative.

A subsequent test could assess clearer presentation of comparable specifications or purchase conditions, with retailer agreement. No deficient page has been observed. Purchase conversion should remain the commercial outcome; cart addition is a supporting signal. This follows the course distinction between KPI activity and the underlying objective. [C2; C5]

19. If the cart-to-purchase comparison also raises concerns

Request delivery cost/timing, stock, payment and error information before attributing the difference to checkout. Cart users are a selected population. Delayed or cross-session purchases can also fall outside the current outcome definition.

Session 3 distinguishes brand-owned from partner-owned touchpoints. LG can support product content, but cannot be assumed to control this unidentified retailer's checkout. Any recommendation should identify which party can implement the change and what cooperation is needed. [C3]

20. If the gap is localised, or does not hold up

The top-product sensitivity already reduces the adjusted gap from -0.827 to -0.431 points. Treat that as a reason to investigate the observed assortment and whether a small set of models drives the story. It is not proof that those products are poor, or that removing them improves sales.

If later checks leave a small or unstable difference, recommend more evidence or no immediate intervention based on this study. A price association alone does not justify a discount. The pricing lessons require demand and contribution evidence that this dataset does not provide. [C16; N17]

21. What a future test proposal would need

Specify one change, audience, channel, responsible party, allocation unit, purchase KPI and observation window. A future visitor-level test may use a different unit from the historical product-consideration analysis; state that explicitly. Returns, margins and customer-experience guardrails would require additional data.

No causal lift, ROI, test duration or required sample size has been established. Current traffic, a worthwhile effect and an implementable design would be needed. A future test is a proposal for better evidence, not an experiment already contained in the CSV.

DELIVERABLES & DECISIONS

What the group will do next

22. The interim presentation

Submit **30420_Interim_Group[N].pdf** by **31 October 2026, 23:59 Milan time**. The PDF has at most five slides, including the cover; one member uploads to Blackboard. Include group and members, brand introduction and rationale, problem, primary RQ and dataset. Proposed methods are optional and a preliminary proposal is acceptable. [G, pp. 3-4]

Slide	Purpose and essential content
1. Brand and context	LG, TV, Samsung, group and members; why this focused project.
2. Managerial problem	Preliminary 2.27% versus 3.13% conversion; the investigation decision.
3. Research question	Primary adjusted comparison, supporting journey diagnostic and scope.
4. Available data	Behavioural file, row definition, cohort counts and historical-proxy limits.
5. Analytical approach	Journey KPIs + Logit; concentration and uncertainty checks. LPM is supporting.

Use the fresh audit to justify feasibility; it does not require presenting every model in the interim. The method story can stay simple: define conversion, adjust for observable price/time differences, and check whether the conclusion depends on a few products. Avoid a slide listing all course techniques.

23. What is settled, and what remains open

The raw-data audit supports the existing TV direction and reproduces the baseline. The group still needs to own and reproduce the analysis, agree the scope and discuss inference and comparability with the instructor. Product concentration, missing specifications and observation-window choices remain material. No final intervention has been selected.

Bring those decisions to the 5 November tutorial, and robustness plus a provisional recommendation to 16 November. Final materials are due 25 November and the class presentation is 26 November. The separate execution plan details the work. Supplied course dates remain subject to Blackboard updates. [G]

Sources and verification scope

[G] 30420_Group_Project_Guidelines.pdf, pp. 1-5. **[D]** DATA_DICTIONARY.md. **[R]** supplied LG_TV_Project_Rationale_and_Plan_EN.pdf, the structural base. **[A]** fresh audit of both supplied CSVs, 25 October 2026: schemas, cohort, descriptions, model fits and stated sensitivities. Logit was fitted independently by Newton maximum likelihood in NumPy; LPM by least squares. Cluster sandwich covariance and a delta-method probability contrast were computed. Counts, baseline probabilities and odds ratio reconcile with [R]. This is not a rerun of the earlier scripts or a claim that every robustness check is complete.

Direct course sources: [C2/C3/C5] Sessions 2, 3 and 5; [N6] session6_kpi_case; [C7] Session7_OLS_Extention_Application; [C8/N9] Session8_STDP_Segmentation and session9_segmentation_case; [C10/N11] Session10_Positioning_Perceptual_Map and session11_positioning_perceptual_map-2; [C12/N15] Session12_Logit and session15_response_models; [C16/N17] Session16_Pricing_Promotion and session17_pricing_case. Lecture files are PDFs; practice files are notebooks.

Internal planning support. The course permits AI for coding, debugging and grammar checking, but not drafting the final report; the group must provide its own interpretation and the required AI-use disclosure. [G, p. 5]